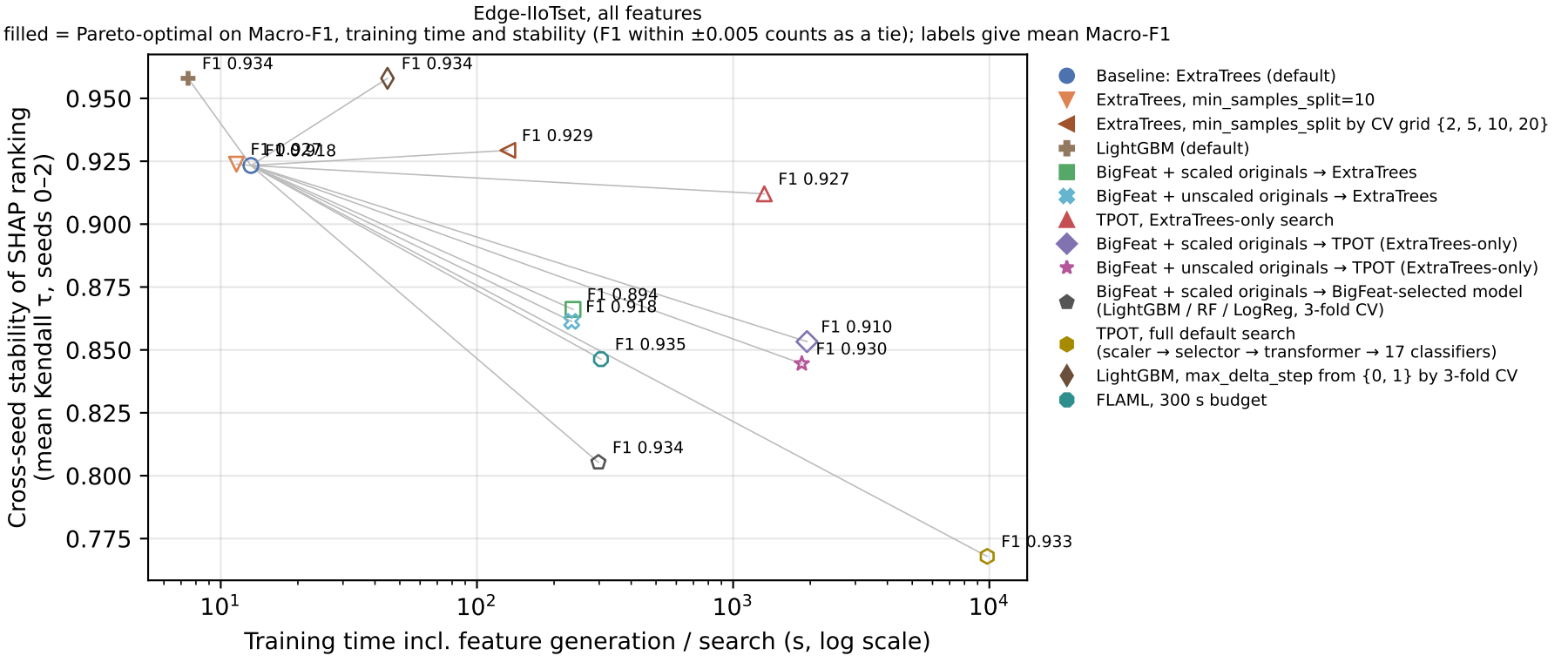
Edge-IIoTset, all features
filled = Pareto-optimal on Macro-F1, training time and stability (F1 within ±0.005 counts as a tie); labels give mean Macro-F1
Cross-seed stability of SHAP ranking
(mean Kendall τ, seeds 0–2)
Training time incl. feature generation / search (s, log scale)
0.950
0.925
0.900
0.875
0.850
0.825
0.800
0.775
10^1
10^2
10^3
10^4
F1 0.934
F1 0.934
F1 0.929
F1 0.927
F1 0.894
F1 0.918
F1 0.910
F1 0.930
F1 0.935
F1 0.934
F1 0.933
Baseline: ExtraTrees (default)
ExtraTrees, min_samples_split=10
ExtraTrees, min_samples_split by CV grid {2, 5, 10, 20}
LightGBM (default)
BigFeat + scaled originals → ExtraTrees
BigFeat + unscaled originals → ExtraTrees
TPOT, ExtraTrees-only search
BigFeat + scaled originals → TPOT (ExtraTrees-only)
BigFeat + unscaled originals → TPOT (ExtraTrees-only)
BigFeat + scaled originals → BigFeat-selected model
(LightGBM / RF / LogReg, 3-fold CV)
TPOT, full default search
(scaler → selector → transformer → 17 classifiers)
LightGBM, max_delta_step from {0, 1} by 3-fold CV
FLAML, 300 s budget